The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction and Architectural Core

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 The 30-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 30}$. For each node i, the vector decomposes into six distinct socio-pragmatic and deontic tensor blocks:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{emp (16)}} \mid \boldsymbol{g}_i^{\text{spatial (2)}} \mid \boldsymbol{p}_i^{\text{ppl (3)}} \mid \boldsymbol{r}_i^{\text{rsa (3)}} \mid \boldsymbol{s}_i^{\text{searle (3)}} \mid \boldsymbol{t}_i^{\text{deontic (3)}} \right] \in \mathbb{R}^{30} \tag{1}$$

3 Homomorphic Multi-Agent Lattices and Fitting Consensus

The architecture unifies the Epistemic Knowledge Lattice $\mathcal{L}_K$ and the Security Flow Lattice $\mathcal{L}_S$ into an identical branching partial-order structure governed by $\nu \in \mathcal{A}$. Parallel worker agents a_1 and a_2 occupy the same tier coordinate, formalizing a non-linear poset governed by the following strict axiomatic constraints:

$$a_3 \left(\alpha_3 \preceq a_2\right), \quad a_3 \left(\alpha_3 \preceq a_1\right), \quad a_2 \preceq a_0, \quad a_1 \preceq a_0 \tag{2}$$

4 Adversarial Defenses and Deep Security Reinforcement Learning

We formalize the automated protection of the multi-agent system (MAS) perimeter as a non-zero-sum Markov Game operating over an advanced windowed token filtering gateway.

4.1 Reduced-Vocabulary Intrusion Detection Interface

To protect the active multi-agent system from open-ended adversarial injection, the LLM user agents do not communicate directly with the symbolic call-stack. Instead, all input prompts are routed through a pre-built Intrusion Detection System (IDS). This gateway processes the incoming strings and projects them into a strictly *reduced vocabulary form* before passing the tokens to the neuro-symbolic (NS) agents. This interface decoupling is maintained as a deliberate research boundary: because the current phase focuses on multi-agent consensus networks, the pre-built IDS acts as an isolated, upstream firewall layer.

4.2 Google Windowed Perplexity Filter Gateway

Within the reduced-vocabulary stream, the NS intrusion detection filter computes the maximum windowed per-token log-likelihood profile over a sliding token window $\mathcal{W}$ of size k to isolate non-human-interpretable adversarial tokens:

$$\text{PPL}_{\max} = \max_{i} \left[\frac{1}{k} \sum_{j=i}^{i+k-1} \log P(\text{token}_j \mid \text{context}) \right] \tag{3}$$

If $\text{PPL}_{\max} < \tau$, where $\tau = -4.0000$ represents the security boundary threshold, an adversarial payload exception is triggered. The engine immediately refuses response execution, and grafts a text-notational Zanzibar relation tuple directly into the decision forest branch leaf nodes:

$$\langle \text{namespace} \rangle : \langle \text{object_id} \rangle \# \langle \text{relation} \rangle @ \langle \text{user_id} \rangle \tag{4}$$

4.3 Symmetric Progression-Based DRL Game Mechanics

The multi-agent system conditions defensive policy vectors by optimizing a fully observable defender reward function r_1 and a partially observable attacker reward function r_2. Let $\mathbb{I}$ represent the indicator function for active system state transitions. The structural rewards scale non-linearly over the attack progression matrix:

$$r_1(h_{ts}) = -\lambda_1 \cdot \mathbb{I}(\text{Exploit}) - \lambda_2 \cdot \mathbb{I}(\text{Impact}) - \lambda_3 \cdot \mathbb{I}(\text{Restore}) \tag{5}$$

$$r_2(h_{t2}) = \lambda_1 \cdot \mathbb{I}(\text{Exploit}) + \lambda_2 \cdot \mathbb{I}(\text{Impact}) \tag{6}$$

where $\lambda_1 = 2000$, $\lambda_2 = 1000$, and $\lambda_3 = 500$. The attacker's blind spot regarding the quality of service (QoS) degradation penalty λ_3 during a network `Restore` operation allows the defender's adaptive policy (π_1) to maintain dominance, minimizing long-term variance and securing the systemic performance space V'.